

PROJECT APEX: MEDICAL CLINIC FIT-OUT

PHASE 3 & 4: SCHEDULING & CONSTRUCTION DELIVERY CONTROLS

Phase 3: Master Schedule & Critical Path Architecture

This section outlines the baseline project schedule mapping out the 24-week delivery sequence. Special emphasis is placed on identifying the **Critical Path Method (CPM)** dependencies where zero total float exists.

Timeline	Work Package Description	Predecessors	Critical Path Status / PM Logic Note
Weeks 1–4	Mobilization & Structural Prep: Site handover, utility isolation, 3D laser-scanning, structural surveys.	None	Standard Baseline Sequence.
Weeks 5–8	Structural Partitions: Metal-stud installation mapping out 6 treatment rooms. Lead-lining walls for radiation shielding in the X-ray suite.	Weeks 1–4	High priority structural interface.
Weeks 9–14	First-Fix MEP Services: Core installation of medical gas pipelines, specialized clinical HVAC ductwork, and primary medical electrical distribution lines.	Weeks 5–8	CRITICAL PATH (● Zero Float). Dry-lining cannot close walls until MEP services are pressure-tested and signed off.
Weeks 15–18	Second-Fix & Hygienic Finishes: Wall closure via cleanable moisture-resistant board, seamless anti-microbial covered vinyl flooring (NHS HTM compliance).	Weeks 9–14	Dependent entirely on first-fix validation.
Weeks 19–22	Equipment Testing & Commissioning: Medical chair installs, HVAC air balancing calibration, medical gas certification, and punch list snagging.	Weeks 15–18	CRITICAL PATH (● Zero Float). Directly linked to clinical licensing and insurance handovers.
Weeks 23–24	Handover & Practical Completion: Delivery of Health & Safety File, O&M Manuals, asset tags, and formal keys-to-client transfer.	Weeks 19–22	Final Delivery Interface.

Phase 4: Construction Execution & Variation Controls (Live Case Scenario)

To demonstrate practical problem-solving using the APM Framework, this live case scenario tracks how an unexpected on-site variable is dynamically resolved through contractual change controls.

Live Construction Event: Discovery of Latent Defects

The Issue (Week 6): During metal-stud partition tracking, structural contractors uncovered unmapped sub-floor wet rot originating from an ancient legacy drainage pipe beneath the designated X-ray room slab. Left untreated, the floor cannot legally support the load of heavy imaging machinery.

Impact Assessment: Initial calculations show a potential 3-week critical path delay to dry-lining and an unmitigated structural remediation cost of **£18,500**.

APM Change Control Resolution Framework

As the Construction Project Manager, the following three-step protocol was immediately executed to defend the client's triple constraints:

1. **Contractual Warning Mechanism:** Issued an immediate formal *Early Warning Notice (EWN)* under the JCT 2024 terms to protect the client legally and logistically. Convened an emergency risk-reduction meeting with the structural engineer within 24 hours.
2. **Schedule Re-sequencing (Mitigating Time):** While the concrete slab was being cut, dried, and chemically treated over 10 days, the dry-lining and partition trades were re-sequenced to focus entirely on the non-impacted zones (reception, staff rooms, and WCs). This parallel processing safely reduced the 3-week critical path delay down to exactly **zero days** of net project drift.
3. **Contingency Drawdown (Mitigating Cost):** Formally approved a **£18,500 Variation Order** funded out of the initial £65,000 project contingency budget. The updated available contingency reserve stands safely at £46,500, preserving the baseline contract target of £650,000.